Sports Science Research Update

July 2023

Connective adaptive resistance exercise (CARE) machines for accentuated eccentric and eccentric-only exercise: Introduction to an emerging concept. Sports Medicine, Ahead of Print: 1-14, 2023.

Incidence of football injuries sustained on artificial turf compared to grass and other playing surfaces: A systematic review and meta-analysis. The Lancet-eClinicalMedicine, 59(101956): 1-13, 2023.

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Load management: What it is and what it is not! Sports Health, Ahead of Print: 1, 2023.

Lower extremity strains in the US National Football League, 2015-2019. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2023.

MRI findings associated with anterior cruciate ligament tears in National Football League athletes. The Orthopaedic Journal of Sports Medicine, 11(6): 1-9, 2023.

Musculoskeletal injury in American football: A bibliometric analysis of the most cited articles. The Orthopaedic Journal of Sports Medicine, 11(6): 1-11, 2023.

On the predictive validity of the National Football League Combine: Does it forecast future success? Journal of Sports Sciences, Ahead of Print: 1-16, 2023.

Primary and secondary risk factors associated with concussion symptom clusters in collegiate athletes: Results from the NCAA-DoD Grand Alliance CARE Consortium. The Orthopaedic Journal of Sports Medicine, 11(4): 1-10, 2023.

Ramping isometrics for accelerated return to play following hamstring tendon repair: A case study. Journal of Elite Sport Performance, 3(1): 1-15, 2023.

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Tackling rugby safety through a collective approach. British Journal of Sports Medicine, Ahead of Print: 1-2, 2023.

The acute demands of repeated-sprint training on physiological, neuromuscular, perceptual and performance outcomes in team sport athletes: A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-32, 2023.

The effects of mental fatigue on sport-specific motor performance among team sport athletes: A systematic scoping review. Frontiers in Psychology, April 11: 1-13, 2023.

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Topical cannabidiol is well tolerated in individuals with a history of elite physical performance and chronic lower extremity pain. Journal of Cannabis Research, 5(11): 1-10, 2023.

Use of publicly obtained data in sports medicine research – A systematic review and bibliometric analysis. The American Journal of Sports Medicine, Ahead of Print: 1-7, 2023.

Why sports should embrace bilateral asymmetry: A narrative review. Symmetry, 14(1993): 1-18, 2023.

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Connective adaptive resistance exercise (CARE) machines for accentuated eccentric and eccentric-only exercise: Introduction to an emerging concept. Sports Medicine, Ahead of Print: 1-14, 2023.

Abstract - Eccentric resistance exercise emphasizes active muscle lengthening against resistance. In the past 15 years, researchers and practitioners have expressed considerable interest in accentuated eccentric (i.e., eccentric overload) and eccentric-only resistance exercise as strategies for enhancing performance and preventing and rehabilitating injuries. However, delivery of eccentric resistance exercise has been challenging because of equipment limitations. Previously, we briefly introduced the concept of connected adaptive resistance exercise (CARE)—the integration of software and hardware to provide a resistance that adjusts in real time and in response to the individual's volitional force within and between repetitions. The aim of the current paper is to expand this discussion and explain the potential for CARE technology to improve the delivery of eccentric resistance exercise in various settings. First, we overview existing resistance exercise equipment and highlight its limitations for delivering eccentric resistance exercise. Second, we describe CARE and explain how it can accomplish accentuated eccentric and eccentric-only resistance exercise in a new way. We supplement this discussion with preliminary data collected with CARE technology in laboratory and non-laboratory environments. Finally, we discuss the potential for CARE technology to deliver eccentric resistance exercise for various purposes, e.g., research studies, rehabilitation programs, and home-based or telehealth interventions. Overall, CARE technology appears to permit completion of eccentric resistance exercise feasibly in both laboratory and non-laboratory environments and thus has implications for researchers and practitioners in the fields of sports medicine, physiotherapy, exercise physiology, and strength and conditioning. Nevertheless, formal investigations into the impact of CARE technology on participation in eccentric resistance exercise and clinical outcomes are still required.

Key Points

- Delivery of eccentric resistance exercise is often unfeasible because of mechanical and practical limitations of free weights, weight stack machines, and plated-loaded machines
- Connected adaptive resistance exercise is the integration of software and hardware to provide a resistance that adjusts in real time and in response to the individual's volitional force within and between repetitions
- Connected adaptive resistance exercise technology can deliver accentuated eccentric and eccentric-only resistance exercise safely and feasibly and thus has the potential to increase

participation in, and dose potency of, eccentric resistance exercise in multiple settings and in various populations

Sports Science Research Update

Incidence of football injuries sustained on artificial turf compared to grass and other playing surfaces: A systematic review and meta-analysis. The Lancet-eClinicalMedicine, 59(101956): 1-13, 2023.

Summary: Background Prior reviews have not conducted statistical synthesis of injury incidence on artificial turf in football. To analyze and compare the incidence of injuries sustained playing football (soccer) on artificial turf compared to grass and other playing surfaces.

Methods: This was a systematic review and meta-analysis. We searched PubMed, Scopus, SPORTDiscus, and Web of Science databases in October 2022 without filters. All observational studies (prospective or retrospective) that analyzed injuries sustained playing football on artificial turf and which included a control group that played on grass or other surface were included. Studies were included if they reported the number of injuries and the exposure time for the playing surfaces. Risk of bias was assessed by Newcastle-Ottawa Scale. A random effects model was used to calculate the pooled incidence rate ratios (IRR) with 95% confidence intervals. Protocol was registered with PROSPERO on October 30th, 2022. Registration number: CRD42022371414.

Findings: We screened 1447 studies, and evaluated 67 full reports, and finally included 22 studies. Risk of bias was a notable issue, as only 5 of the 22 studies adjusted their analysis for potential confounders. Men (11 studies: IRR 0.82, CI 0.72–0.94) and women (5 studies: IRR 0.83, CI 0.76–0.91) had lower injury incidence on artificial turf. Professional players had a lower incidence of injury (8 studies: IRR 0.79, CI 0.70–0.90) on artificial turf, whereas there was no evidence of differences in the incidence of injury in amateur players (8 studies: IRR 0.91, CI 0.77–1.09). The incidence of pelvis/thigh (10 studies: IRR 0.72, CI 0.57–0.90), and knee injuries (14 studies: IRR 0.77, CI 0.64–0.92) were lower on artificial turf.

Interpretation: The overall incidence of football injuries is lower on artificial turf than on grass. Based on these findings, the risk of injury can't be used as an argument against artificial turf when considering the optimal playing surface for football.

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Investigating the effect of return to play timing after injury on performance: Does the analysis answer the research objective? Sports Medicine, Ahead of Print: 1-10, 2023.

Abstract - Return-to-play decision making should be based on all the advantages and disadvantages of return to play for athletes, not just the risk of injury. For competitive athletes, this includes the effect of early versus delayed return to sport on performance. In this paper, we address the questions "How can I estimate the effect of injury on the individual's performance at return to play?" and "What is the effect of delaying return to sport on the individual's performance?". To address these questions, we describe (1) some foundational concepts, design and analytical challenges related to estimating the causal effect of return to play timing on performance in the athlete, (2) additional challenges if one is interested in the effects of delaying return to play and (3) differences when the questions relate to the team's performance. Although the analytical strategies described appear complicated, coaches and athletes make these judgements informally every day without explicitly stating their assumptions. Using a formal approach should help analysts provide the most valid answers to the questions asked by athletes and coaches. In brief, the choice of a comparison group depends on the research question and requires that one consider the hypothetical performance trajectory of the athlete had they never been injured. Thus, the optimal comparison group depends on the shape of the expected trajectory and the specific research question being asked.

Key Points

- Choosing the comparison group for an observational study investigating the effect of injury on performance depends on the specific research question
- Estimating the causal effect of injury on performance requires comparing what happened to performance after injury (which is observed and "factual") to what would have happened to performance if participants had not been injured (which is not observed and "counterfactual"

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Load management: What it is and what it is not! Sports Health, Ahead of Print: 1, 2023.

For sport spectators who regularly see their favorite player sitting on the bench, it would be easy to conclude that "load management" predominantly involves greater rest for players, reduced game time, and an opportunity to remove players from training and/or competition. However, while adequate recovery is important to ensure optimal adaptations to a given training stimulus, "recovery" is only one of many training principles that sports medicine and performance staff use to develop physical qualities, minimize injury risk, and ultimately enhance performance. These well-established training principles also include progressive overload, specificity, individuality, variation, and reversibility.

Graded exposure to stress (ie, load) is a fundamental component of evidence-based training programs. In order to improve capacity, the training load applied must exceed the athlete's current capacity, but not be so great that it results in tissue damage.8 Progressive overload involves the systematic application of training stress and is the cornerstone of rehabilitation, return-to-sport, and performance training programs.3 Closely aligned with progressive overload is the principle of specificity, as early as the 1940s, researchers demonstrated that the biochemical response of a tissue depended on the type of training that was performed.7,9 These early studies played a significant role in shaping our understanding of the "needs analysis" for athletes.1

While it is acknowledged that athletes need to be exposed to high training loads to tolerate the high-intensity demands of competition,2 these high training loads need to be interspersed with low-intensity training sessions. This polarized approach to training describes the variation principle and has been used as an effective means of enhancing performance in both sprinters and endurance athletes.5,6 The principle of individuality states that athletes will respond differently to a given training stimulus. This explains, at least in part, why some athletes can tolerate large training loads (eg, think playing minutes) and others cannot. Although load progressions are important to improve capacity, on occasions the training loads of athletes may need to be sensibly regressed (or greater recovery time provided) to minimize negative training adaptations such as excessive fatigue, pain, reductions in performance, and increased reinjury risk. The challenge facing all sports medicine and performance practitioners is ensuring that these reductions in training load do not occur for extended periods of time, otherwise local tissue capacity and sport-specific capacity may be reduced.4 These reductions in capacity that occur during prolonged decreases in training load are referred to as the reversibility principle. Essentially, this principle states that "if you don't use it, you'll lose it."

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In recognition of the importance of “load management” to professional athletes, military personnel, artistic performers (eg, circus and dancers), adolescent and master athletes, and the general population, and to clarify what it is, and what it is not, Sports Health is devoting a special issue to the topic. Papers that have a clear applied outcome and encompass one or more of the training principles under the “load management” umbrella will be considered for peer review. We invite researchers and practitioners to submit their papers by December 1, 2023. Please email christine@aossm.org to be included in this issue and share the title of the paper you plan to submit. We look forward to your contribution to this leading-edge special issue.

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Sports Science Research Update

Lower extremity strains in the US National Football League, 2015-2019. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2023.

Background: Lower extremity (LEX) strains, including hamstring, quadriceps, adductor, and calf strains, are among the most common injuries in sports. These injuries lead to high burden, resulting in significant missed participation time.

Purpose: To describe the incidence of LEX strains in professional American football.

Study Design: Descriptive epidemiology study.

Methods: This study included all players who played in 1 National Football League (NFL) game or sustained a LEX strain during participation in the 2015-2019 seasons. LEX strain frequency was calculated by setting (game, practice, conditioning), timing in season (offseason, preseason, regular season, postseason), and roster position. Game incidence rates were calculated for season, roster position, and play type. LEX strains were identified in the standardized leaguewide electronic health record (n = 32 teams).

Results: Across 5 years, 5780 LEX strains were reported among 2769 players (1-year risk, 26.7%; 95% CI, 26.0%-27.3%); 69% (n = 4015) resulted in time loss. Among all LEX strains, 54.7% were hamstring (n = 3163), 24.1% adductor (n = 1393), 12.6% calf (n = 728), 8.3% quadriceps (n = 477), and 0.3% multiple muscle groups (n = 19). Most were reported during preseason practices (n = 1076; 27%) and regular season games (n = 1060; 26%). The 2-week period of training camp practices comprised 19% of all time-loss strains. Among game injuries, preseason games had the highest rate of LEX strain (2.9/10,000 player-plays; 95% CI, 2.6-3.2). Defensive secondary players accounted for the highest proportion of time-loss LEX strains (27%; n = 1082). In games, punt plays had nearly twice the injury rate of kickoff plays (14.9/1000 plays [95% CI, 13.1-17.0] vs 7.5/1000 plays [95% CI, 6.2-8.9], respectively) and .3 times the rate of pass plays (4.3/1000 plays; 95% CI, 4.0-4.7) and run plays (2.6/1000 plays; 95% CI, 2.3-2.9). In aggregate, LEX strains led to an estimated 16,748 participation days missed each year and a median 12 days missed per injury.

Conclusion: LEX strains affected 1 in 4 NFL players each year, resulting in a high burden of injury in terms of time lost from practice and competition. Safe return to the NFL season during training camp and reduction of injuries during regular season games are key focuses for future injury reduction.

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MRI findings associated with anterior cruciate ligament tears in National Football League athletes. The Orthopaedic Journal of Sports Medicine, 11(6): 1-9, 2023.

Background: Anterior cruciate ligament (ACL) tears are a high-frequency injury requiring a lengthy recovery in professional American football players. Concomitant pathology associated with ACL tears as identified on magnetic resonance imaging (MRI) is not well understood in these athletes.

Purpose: To describe the MRI findings of concomitant injuries associated with ACL tears among athletes in the National Football League (NFL).

Study Design: Cross-sectional study; Level of evidence, 3.

Methods: Of 314 ACL injuries in NFL athletes from 2015 through 2019, 191 complete MRI scans from the time of primary ACL injury were identified and reviewed by 2 fellowship-trained musculoskeletal radiologists. Data were collected on ACL tear type and location, as well as presence and location of bone bruises, meniscal tears, articular cartilage pathology, and concomitant ligament pathology. Mechanism data from video review were linked with imaging data to assess association between injury mechanism (contact vs noncontact) and presence of concomitant pathology.

Results: Bone bruises were evident in 94.8% of ACL tears in this cohort, most often in the lateral tibial plateau (81%). Meniscal, additional ligamentous, and/or cartilage injury was present in 89% of these knees. Meniscal tears were present in 70% of knees, lateral (59%) more than medial (41%). Additional ligamentous injury was present in 71% of all MRI scans, more often a grade 1/2 sprain (67%) rather than a grade 3 tear (33%), and most often involving the medial collateral ligament (MCL) (57%) and least often the posterior cruciate ligament (10%). Chondral damage was evident in 49% of all MRI scans, with ≥ 1 full-thickness defect in 25% of all MRI scans, most often lateral. Most (79%) ACL tears did not involve direct contact to the injured lower extremity. Direct contact injuries (21%) were more likely to have a concomitant MCL tear and/or medial patellofemoral ligament injury and less likely to have a medial meniscal tear.

Conclusion: ACL tears were rarely isolated injuries in this cohort of professional American football athletes. Bone bruises were almost always present, and additional meniscal, ligamentous, and chondral injuries were also common. MRI findings varied by injury mechanism.

Sports Science Research Update

Musculoskeletal injury in American football: A bibliometric analysis of the most cited articles. The Orthopaedic Journal of Sports Medicine, 11(6): 1-11, 2023.

Background: Textbook knowledge and clinical dogma are often insufficient for effective evidence-based decision making when treating musculoskeletal injuries in American football players, given the variability in presentation and outcomes across different sports and different levels of competition. Key evidence can be drawn directly from high-quality published articles to make the appropriate decisions and recommendations for each athlete's unique situation.

Purpose: To identify and analyze the 50 most cited articles related to football-related musculoskeletal injury to provide an efficient tool in the arsenal of trainees, researchers, and evidence-based practitioners alike.

Study Design: Cross-sectional study.

Methods: The ISI Web of Science and SCOPUS databases were queried for articles pertaining to musculoskeletal injury in American football. For each of the top 50 most cited articles, bibliometric elements were evaluated: citation count and density, decade of publication, journal, country, multiple publications by the same first author or senior author, article content (topic, injury area), and level of evidence (LOE).

Results: The mean 6 SD number of citations was 102.76 6 37.11; the most cited article, with 227 citations, was ''Syndesmotic Ankle Sprains'' published in 1991 by Boytim et al. Several authors served as a first or senior author on .1 publication, including J.S. Torg (n = 6), J.P. Bradley (n = 4), and J.W. Powell (n = 4). The American Journal of Sports Medicine published the majority of the 50 most cited articles (n = 31). A total of 29 articles discussed lower extremity injuries, while only 4 discussed upper extremity injuries. The majority of the articles (n = 28) had an LOE of 4, with only 1 article having an LOE of 1. The articles with an LOE of 3 had the highest mean citation number (133.67 6 55.23; F = 4.02; P = .05).

Conclusion: The results of this study highlight the need for more prospective research surrounding the management of football-related injury. The low overall number of articles on upper extremity injury (n = 4) also highlights an area for further research.

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On the predictive validity of the National Football League Combine: Does it forecast future success? Journal of Sports Sciences, Ahead of Print: 1-16, 2023.

Abstract - The National Football League (NFL) Combine provides NFL teams the ability to assess prospective athletes' medical histories and physical and psychological abilities. Using this information, NFL personnel must then decide whether an athlete is a good fit to their team. Given the combine's 40-year history and the availability of peer reviewed articles on the function and efficacy of the combine, the purpose of this systematic review was to synthesize the literature evaluating the predictive validity of the combine according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses). Full-text, peer-reviewed articles containing information relevant to the NFL combine and at least one measure of future success were retained. The search yielded 1954 articles and after screening, 68 articles remained. These remaining articles focused on measures of success pertaining to (a) medical testing and future performance (n = 25), (b) combine tests (n = 12), (c) draft position (n = 10), (d) draft and future performance (n = 8), (e) medical testing (n = 7), (f) career longevity (n = 3), (g) draft and salary (n = 2), and (h) playing performance (n = 1). Due to the mixed results of combine measures on future success, this review highlights the need for more research to investigate the combine's influence on long-term performance and success.

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Primary and secondary risk factors associated with concussion symptom clusters in collegiate athletes: Results from the NCAA-DoD Grand Alliance CARE Consortium. The Orthopaedic Journal of Sports Medicine, 11(4): 1-10, 2023.

Background: There is a broad and diverse range of symptoms after a concussion, from irritability to nausea. This heterogeneity of symptoms is a challenge for clinicians managing the different presentations among injuries. Prior research has investigated the structure of postconcussive symptoms to determine if they can be grouped into clusters of related symptoms.

Purpose/Hypothesis: The purpose of this study was to identify symptom clusters during the acute phase after a sports-related concussion using exploratory factor analysis and to understand the relationship between risk factors for postconcussion symptoms (ie, demographics, injury characteristics, mental health, and sleep qualities) and different symptom clusters. We hypothesized that certain factors would be predictive of specific symptom clusters.

Study Design: Cross-sectional study; Level of evidence, 3.

Methods: Collegiate athletes (N = 1104) from the Concussion, Assessment, Research, and Education (CARE) Consortium completed the Sport Concussion Assessment Tool–Third Edition symptom assessment tool 24 to 48 hours after concussion. Exploratory factor analysis was conducted on the symptom evaluation to determine symptom clusters 24 to 48 hours after concussion. Regression analysis was used to examine the effects of pre- and postinjury characteristics.

Results: Exploratory factor analysis revealed a 4-cluster structure for acute postconcussive symptoms that explained 62% of the variance in symptom reporting: vestibular-cognitive, migrainous, cognitive fatigue, and affective. Delayed reporting, less sleep before assessment, female sex, and being hurt outside of competition (during practice/training) was correlated with increased symptoms for 4 symptom clusters. Depression predicted higher vestibular-cognitive and affective symptoms. Amnesia was correlated with higher vestibular-cognitive and migrainous symptoms, whereas migraine history was associated with more migrainous and affective symptoms.

Conclusion: Symptoms can be grouped into 1 of 4 distinct clusters. Certain variables were associated with increased symptoms across multiple clusters and may be indicative of greater injury severity. Other factors (ie, migraine history, depression, amnesia) were associated with a more specific symptom presentation and may be mechanistically related to concussion outcomes and biological markers.

Sports Science Research Update

Ramping isometrics for accelerated return to play following hamstring tendon repair: A case study. Journal of Elite Sport Performance, 3(1): 1-15, 2023.

Abstract - Semimembranosus injuries are becoming increasingly common in rugby union, often via an overstretch mechanism, particularly from the jackal action in deep hip flexion with simultaneous knee extension. This injury often requires surgical repair, leading to an estimated return to play (RTP) time of 16 weeks. This case study documents the RTP of an elite international rugby union player following a combined semimembranosus, semitendinosus, and biceps femoris injury, with surgical reconstruction of the retracted semimembranosus tendon. Using an early low-jerk ramping isometric protocol, the player returned to team training at 10 weeks with RTP at exactly 12 weeks post-operation. He continued to play at international and club level for the remaining 2 seasons, without re-injury. This case study demonstrates how the early introduction of a ramped isometric protocol can be safe for muscle-tendon injuries and may improve expected recovery rates.

Sports Science Research Update

Risk of upper extremity musculoskeletal injury within the first year after a concussion. The Orthopaedic Journal of Sports Medicine, 11(5): 1-9, 2023.

Background: Emerging evidence suggests that athletes and military personnel are at increased risk for lower extremity musculoskeletal injury after a concussion; however, the association between concussion and subsequent upper extremity (UE) musculoskeletal injury is unknown.

Purpose: To prospectively examine the association between concussion and UE musculoskeletal injury risk within the first year after returning to unrestricted activity.

Study Design: Cohort study; Level of evidence, 3.

Methods: A total of 316 cases of concussion 42% (132/316 women) were observed among 5660 Concussion Assessment, Research and Education Consortium participants at the United States Military Academy from May 2015 to June 2018. Active injury surveillance within the cohort was conducted for 12 months after unrestricted return to activity to identify any incident cases of acute UE musculoskeletal injury. Injury surveillance during the follow-up period was also conducted for nonconcussed controls who were matched by sex and competitive sport level. Univariate and multivariable Cox proportional hazards regression models were used to estimate hazard ratios between concussed cases and nonconcussed controls for time to UE musculoskeletal injury.

Results: During the surveillance period, 19.3% of concussed cases and 9.2% of nonconcussed controls sustained a UE injury. In the univariate model, concussed cases were 2.25 times (95% CI, 1.45-3.51) more likely to sustain a UE injury during the 12-month follow-up period when compared with the nonconcussed controls. In the multivariable model, adjusted for history of concussion, sport level, somatization, and history of UE injury, concussed cases were 1.84 times (95% CI, 1.10-3.07) more likely to sustain a UE injury during the surveillance period compared with nonconcussed controls. Sport level remained an independent risk factor for UE musculoskeletal injury; however, concussion history, somatization, and history of UE injury were not independent risk factors.

Conclusion: Concussed cases were more than twice as likely to sustain an acute UE musculoskeletal injury within the first 12 months after unrestricted return to activity when compared with nonconcussed controls. The higher hazard of injury remained in the concussed group after adjusting for other potential risk factors.

Sports Science Research Update

Tackling rugby safety through a collective approach. British Journal of Sports Medicine, Ahead of Print: 1-2, 2023.

COMMITMENT TO SEEKING AN EVIDENCE-INFORMED APPROACH: When it comes to player welfare, Rugby Union governing bodies have committed to seeking and funding an evidence-informed approach. This involves using research to make informed decisions about policy, laws and injury prevention programs. Over the last decade, a growing body of research has informed player safety, for example, modifications to scrum laws to reduce catastrophic head and neck injuries.[1] However, major gaps remain, including tackle research focusing on the women's game.[2]

A SOCIOECOLOGICAL PERSPECTIVE: It is well understood that player welfare, specifically injury prevention, is a complex issue. To effectively address these complexities and make a long-term impact requires a dynamic socioecological approach.[3 4] An athlete operates within a socioecological structure (individual, interpersonal, organizational, community) that is influenced by a web of inter-related factors and actors, both of which change over time and/or when a factor/actor is modified (figure 1). Typically, injury prevention research will identify player level factors that influence injury risk (risk factors), and aim to modify these factors through behavior change interventions. However, the socioecological view emphasizes understanding contextual factors influencing implementation of such modifications. For example, if the behavior intervention is a training program, how much time is available to implement the training program? Or, is the training program appropriate for all sexes?

REDUCING CONCUSSION RISK DURING THE TACKLE: Reducing concussion risk during the tackle, at all levels, is a top priority for rugby stakeholders and governing bodies. Video analyses examining head impacts and resulting concussions have identified that elements of players' tackling technique are major risk factors. For both effective and safe tackling, the optimal techniques are to maintain a low body position or dip from an upright to low body position to contact the ball-carrier at their center of gravity (mid-torso area). Tackles causing head injuries differ, with upright tackler body positions and head-to-head and head-to-shoulder contact found to be over-represented in injurious tackles.[5]

In view of the concussion risk associated with direct head contacts caused in part by an upright body position,[5] World Rugby (WR) has applied a regulatory approach to change players' tackling behavior through more strict and consistent refereeing of high tackles.[6] WR and national governing bodies have also trialed lowering the height of a legal tackle from the shoulder to the armpit in an effort to 'nudge' players to make contact with the ball-carrier mid-torso.[7 8] France, New Zealand and England have also announced they will implement lowering the legal tackle height in community rugby. These

tackle law changes are related to the maximum height of a legal tackle for officiating purposes, and reinforce the optimal techniques recommended for a safe and effective tackle, that is, contacting the ball-carrier at their center of gravity. [1] Therefore, the change in the maximum height for a legal tackle strengthens our current coaching of proper tackling technique. Changing laws and policies is a 'passive' measure on the passive–active injury prevention and intervention continuum.[9] In general, passive measures have a broader impact as they relate to the whole sport. In contrast, active measures require players, or any party directly responsible for player's safety, to deliberately be involved in protecting the player. Being actively involved also enhances the individual's ability to perform the desired behavior change, that is, self-efficacy, which is a key determinant of behavior change.[10] Integrating passive and active behavior change measures is thought to be most effective in sustainable injury prevention.[9]

Upskilling to play with the laws: To gain self-efficacy in reducing concussion risk in tackles, players must have the ability to repeatedly meet the recognized safety requirement of contacting the ball-carrier at the mid-torso, a skill that requires physical, technical, tactical and psychological proficiency and capacity. A large body of literature describes how to optimize training for skill learning, maintenance, transfer and capacity building. This literature has only recently been applied to tackle training.[11] The interpersonal actor in the socioecological view of preventing injury prevention in team sports is the coach, the target of behavior change for national injury prevention programs such as BokSmart.[12] While research on the specific elements of how to best test, design, provide feedback and evaluate the tackle during training is emerging, we need to acknowledge the complexities of coaching, and recognize that the main outcome measure of a player's tackle ability is performance. Thus, any measure (active or passive) to reduce concussion risk must consider the impact on the player's tackle performance, and accept that any safety measure that impairs performance is unlikely to be implemented.

COLLECTIVE APPROACH: Further, we must appreciate that research alone may be unable to meet the urgent time frames for optimizing tackle safety through skill training. Also, the effectiveness and sustainability of a safe tackle-skill program may be reliant on elements of coaching ('the art of coaching') that cannot necessarily be assessed using a traditional research design. In rugby, and specifically for tackle safety, the need for engagement, collaboration and collective action between stakeholders at all socioecological levels has never been greater— this includes players, coaches, parents, clubs, administrators, governing bodies, media and match officials (figure 1). The necessary next step in this collective action towards improving tackle safety is to bring these stakeholders, along with researchers and clinicians, together to create a specialist tackle working group. The mandate of this working group would be to identify the most important focus areas and research questions to inform best practice and

policy, and then manage and oversee the research process. A key part of this mandate will be to ensure that all research is effectively translated and disseminated to the wider rugby community.

In conclusion, reducing concussion risk during the tackle is complex and requires a dynamic socioecological perspective to integrate active and passive measures to change player behavior. The behavior change required to safely and effectively contest the tackle is a skill, and multistakeholder engagement, collaboration and collective action between players, coaches, parents, clubs, administrators, governing bodies, media and match officials is needed to effectively reduce the risk of concussion in the tackle.

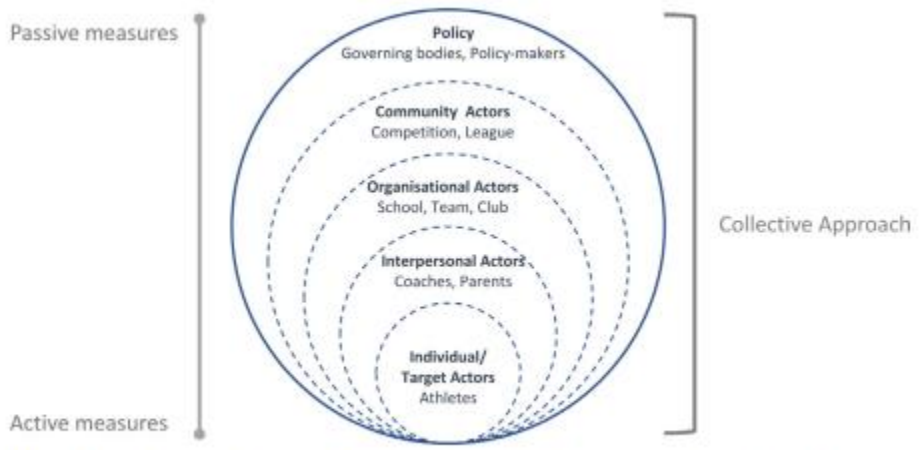

Figure 1 The collective approach of the different actors in the socioecological view for injury prevention in sport along the passive–active intervention continuum. Adapted from [349]

Sports Science Research Update

The acute demands of repeated-sprint training on physiological, neuromuscular, perceptual and performance outcomes in team sport athletes: A systematic review and meta-analysis. Sports Medicine, Ahead of Print: 1-32, 2023.

Background: Repeated-sprint training (RST) involves maximal-effort, short-duration sprints (≤10 s) interspersed with brief recovery periods (≤60 s). Knowledge about the acute demands of RST and the influence of programming variables has implications for training prescription.

Objectives: To investigate the physiological, neuromuscular, perceptual and performance demands of RST, while also examining the moderating effects of programming variables (sprint modality, number of repetitions per set, sprint repetition distance, inter-repetition rest modality and inter-repetition rest duration) on these outcomes.

Methods: The databases Pubmed, SPORTDiscus, MEDLINE and Scopus were searched for original research articles investigating overground running RST in team sport athletes≥16 years. Eligible data were analyzed using multi-level mixed effects meta-analysis, with meta-regression performed on outcomes with ~50 samples (10 per moderator) to examine the influence

of programming factors. Effects were evaluated based on coverage of their confidence (compatibility) limits (CL) against elected thresholds of practical importance.

Results: From 908 data samples nested within 176 studies eligible for meta-analysis, the pooled effects (±90% CL) of RST were as follows: average heart rate (HRavg) of 163 ±9 bpm, peak heart rate (HRpeak) of 182 ±3 bpm, average oxygen consumption of 42.4 ±10.1 mL·kg−1·min−1, end-set blood lactate concentration (B[La]) of 10.7 ±0.6 mmol·L−1, deciMax session ratings of perceived exertion (sRPE) of 6.5 ±0.5 au, average sprint time (Savg) of 5.57 ±0.26 s, best sprint time (Sbest) of 5.52 ±0.27 s and percentage sprint decrement (Sdec) of 5.0±0.3%. When compared with a reference protocol of 6×30 m straight-line sprints with 20 s passive inter-repetition rest, shuttle-based sprints were associated with a substantial increase in repetition time (Savg: 1.42 ±0.11 s, Sbest: 1.55 ±0.13 s), whereas the effect on sRPE was trivial (0.6±0.9 au). Performing two more repetitions per set had a trivial effect on HRpeak (0.8±1.0 bpm), B[La] (0.3±0.2 mmol·L−1), sRPE (0.2±0.2 au), Savg (0.01±0.03) and Sdec (0.4;±0.2%). Sprinting 10 m further per repetition was associated with a substantial increase in B[La] (2.7;±0.7 mmol·L−1) and Sdec (1.7±0.4%), whereas the effect on sRPE was trivial (0.7±0.6). Resting for 10 s longer between repetitions was associated with a substantial reduction in B[La] (−1.1±0.5 mmol·L−1), Savg (−0.09±0.06 s) and Sdec (−1.4±0.4%), while the effects on HRpeak (−0.7±1.8 bpm) and sRPE (−0.5±0.5 au) were trivial. All other moderating effects were compatible with both trivial and substantial effects [i.e.

equal coverage of the confidence interval (CI) across a trivial and a substantial region in only one direction], or inconclusive (i.e. the CI spanned across substantial and trivial regions in both positive and negative directions).

Conclusions: The physiological, neuromuscular, perceptual and performance demands of RST are substantial, with some of these outcomes moderated by the manipulation of programming variables. To amplify physiological demands and performance decrement, longer sprint distances (>30 m) and shorter, inter-repetition rest (≤20 s) are recommended. Alternatively, to mitigate fatigue and enhance acute sprint performance, shorter sprint distances (e.g. 15–25 m) with longer, passive interrepetition rest (≥30 s) are recommended.

Key Points

- The most common RST set configuration is 6×30 m straight-line sprints with 20 s of passive inter-repetition rest.
- The reference estimates for HRavg (90% HR_{max}), VO2avg(~70–80% $VO2_{max}$) and B[La] (10.8 mmol·L−1) demonstrate the substantial physiological demands of RST in team sport athletes. Associated prediction intervals for these estimates suggest that most of these demands are consistently substantial across many RST protocols, sports and athlete characteristics.
- Shorter inter-repetition rest periods (≤20 s) and longer repetition distances (>30 m) amplify physiological demands and cause greater inter-set reductions in sprint performance (i.e. performance fatigue). Inversely, longer inter-repetition rest periods (≥30 s) and shorter repetition distances (≤20 m) enhance acute sprint performance and reduce the physiological demands.
- Shuttle-based protocols are associated with slower repetition times, likely due to the added change-of-direction component, but may reduce sprint decrement. The effect of shuttle versus straight-line RST protocols on physiological and perceptual outcomes remains inconclusive.
- Performing two less repetitions per set (e.g. four as opposed to six repetitions) maintains the perceptual, performance and physiological demands of RST.
- The findings from our investigation provide practitioners with the expected demands of RST and can be used to help optimize training prescription through the manipulation of programming variables.

Sports Science Research Update

The effects of mental fatigue on sport-specific motor performance among team sport athletes: A systematic scoping review. Frontiers in Psychology, April 11: 1-13, 2023.

Background: The psychobiological state known as mental fatigue (MF) is by engaging in mentally taxing activities for an extended period, which is typically found in team sports, of the high cognitive demand and unpredictable environment. It increases the perception of effort and influences executive functions, impairing sport-specific performance in athletes. However, the consequences of MF on sport-specific motor performance (SSMP) among athletes in team sports remain unclear.

Objective: This scoping review seeks to find and map research publications that investigate the effect of MF on SSMP in team sports.

Methods: Web of Science, Scopus, and PubMed were searched as the main databases, and CENTRAL, Psychology, and Behavioral Sciences Collection, SPORTDicus obtained from EBSCOhost, as well as gray literature was searched for relevant literature and Google Scholar. Cognitive tasks before the SSMP exam are the focus of the selected literature on mental exhaustion. Only experiments testing mental and non-mental exhaustion were chosen.

Results: Twelve studies fulfill the requirement of selection criteria. SSMP in team sports, including soccer, basketball, cricket, and Australian football mainly is examined as physical and technical performance. More specifically, MF significantly influenced physical performance measured as intermittent endurance and total distance ($P < 0.05$), while data was inclusive when assess in an ecological setting (e.g., small-sided game) ($P > 0.05$). Technical performance was mainly measured as ball loss, errors in passing and shooting, interception, and successful tackle and showed a dramatic impairment ($P < 0.05$). The decline of physical activity is relevant with higher level PRE, while decreased technical performance is related to impaired attention resources shown as visual perceptual.

Conclusion: MF adversely influences SSMP in team sports. The most relevant theory for future study to examine the impacts of MF on team-sport athletes could be the psychological model of exercise and its potential extension on attention resources, rather than the traditional “catastrophe” theory.

Sports Science Research Update

The Trade Secret Taboo: Opens science methods are required to improve prediction models in sports medicine and performance. Sports Medicine, Ahead of Print: 1-9, 2023.

Abstract - Clinical prediction models in sports medicine that utilize regression or machine learning techniques have become more widely published, used, and disseminated. However, these models are typically characterized by poor methodology and incomplete reporting, and an inadequate evaluation of performance, leading to unreliable predictions and weak clinical utility within their intended sport population. Before implementation in practice, models require a thorough evaluation. Strong replicable methods and transparency reporting allow practitioners and researchers to make independent judgments as to the model's validity, performance, clinical usefulness, and confidence it will do no harm. However, this is not reflected in the sports medicine literature. As shown in a recent systematic review of models for predicting sports injury models, most were typically characterized by poor methodology, incomplete reporting, and inadequate performance evaluation. Because of constraints imposed by data from individual teams, the development of accurate, reliable, and useful models is highly reliant on external validation. However, a barrier to collaboration is a desire to maintain a competitive advantage; a team's proprietary information is often perceived as high value, and so these 'trade secrets' are frequently guarded. These 'trade secrets' also apply to commercially available models, as developers are unwilling to share proprietary (and potentially profitable) development and validation information. In this Current Opinion, we: (1) argue that open science is essential for improving sport prediction models and (2) critically examine sport prediction models for open science practices.

Key Points

- In a recent systematic review of sports injury prediction models, only one of the 30 studies (3%) included in the review made their protocol and analysis plan publicly available.
- In a secondary data extraction for this systematic review, none of the 30 studies made their data or code or data publicly available in an open access repository.
- Fostering collaboration between sports organizations and unlocking 'trade secrets' through available scientific methods can potentially improve athlete care.

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Sports Science Research Update

Topical cannabidiol is well tolerated in individuals with a history of elite physical performance and chronic lower extremity pain. Journal of Cannabis Research, 5(11): 1-10, 2023.

Introduction: Cannabidiol (CBD) is a potential therapeutic for pain management. Yet, there exists a dearth of studies of its tolerability and efficacy, especially in special populations. Former elite athletes are a special population both susceptible to chronic pain and also highly trained and attuned to assess medication tolerability concerns. The purpose of the present open-label pilot study was to assess the tolerability of CBD in this population.

Materials and methods: Retrospective analysis was conducted in deidentified data from 20 individuals who were all previously professional athletes in US/American football, track and field, or basketball, with careers ranging from 4 to 10 years. Participants received topical CBD (10 mg twice daily by controlled dispenser) for chronic pain resulting from acute lower extremity injuries. Assessments of tolerability and secondary analyses of pain, pain-related disability, and activities of daily living were collected by self-report over the 6-week study period. Data were analyzed by descriptive statistics, pairwise t-test, and linear regression.

Results: Seventy percent of participants completed the study. Of the individuals who completed the study, 50% reported minor adverse effects, none of which required medical attention, and 50% did not report any adverse effects. The mostly commonly reported effects were skin dryness (43% of study completers) and skin rash (21% of study completers), which rapidly resolved. There was a significant improvement in self-reported pain levels (intake mean 3.5±0.29; exit mean 1.7±0.23; $P<0.001$) and pain-related disability, including family and home responsibilities, life support activities, occupational activities, recreational activities, self-care, sexual function, and social activities (all $P<0.001$).

Discussion: To the best of our knowledge, this is the first study to assess CBD treatment in elite athletes, who are disproportionally susceptible to disabling injuries. Topical administration of CBD was tolerated well by this population and resulted in only minor adverse effects. As elite athletes are trained and attuned to assess their own bodies due to their professional lives, this population is likely to detect tolerability concerns. However, this study was limited to a convenience sample and self-reported data. These pilot findings warrant further study of topical CBD in randomized and controlled studies of elite athletes.

Sports Science Research Update

Use of publicly obtained data in sports medicine research – A systematic review and bibliometric analysis. The American Journal of Sports Medicine, Ahead of Print: 1-7, 2023.

Background: Publicly obtained data (POD) have recently been utilized frequently by sports medicine researchers to describe injury patterns, risk factors, and outcomes in elite athletes. The relative ease of this type of research that is based solely on inter net and media sources has resulted in a near exponential increase in the number of these POD studies.

Purpose: To systematically review the sports medicine literature for studies based solely on POD.

Study Design: Systematic review and bibliometric analysis; Level of evidence, 4.

Methods: A systematic review of POD studies published since 2000 was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Studies of interest were those relying on publicly available injury reports or online media for data acquisition in collegiate, semiprofessional, and professional athletes.

Results: There were 209 POD studies published between 2000 and 2022, with 173 (82.8%) of these studies published after 2016. Studies were published most frequently on athletes participating in North American professional leagues: National Football League (n = 69 [28.4%]), Major League Baseball (n = 56 [23.0%]), National Basketball Association (n = 37 [15.2%]), and National Hockey League (n = 33 [13.6%]). The most common injuries assessed were head injuries/concussions (n = 43 [21.1%]), anterior cruciate ligament injuries (n = 33 [16.2%]), and ulnar collateral ligament injuries (n = 23 [11.3%]). One-quarter of the studies (n = 53 [25.4%]) reported only 1 POD source, and 1 study (0.5%) reported no source. Additionally, 65 studies (31.1%) listed nonspecific POD resources or solely cited previous literature to describe the POD search methodology and data acquisition.

Conclusion: POD studies are exponentially increasing in number, particularly across major North American professional sports leagues, with significant variability in the injury of interest, search methodology, and number of data sources. The accuracy of the conclusions reached based on the POD methodology appears highly variable. Given the potential impact of these publications as both contributors to current knowledge and drivers of future research, the sports medicine community should be aware of the inherent biases and limitations of POD injury studies.

Sports Science Research Update

Why sports should embrace bilateral asymmetry: A narrative review. Symmetry, 14(1993): 1-18, 2023.

Background: Asymmetry is ubiquitous in nature and humans have well-established bilateral asymmetries in their structures and functions. However, there are (mostly unsubstantiated) claims that bilateral asymmetries may impair sports performance or increase injury risk.

Objective: To critically review the evidence of the occurrence and effects of asymmetry and sports performance.

Development: Asymmetry is prevalent across several sports regardless of age, gender, or competitive level, and can be verified even in apparently symmetric actions (e.g., running and rowing). Assessments of bilateral asymmetries are highly task-, metric-, individual-, and sport-specific; fluctuate significantly in time (in magnitude and, more importantly, in direction); and tend to be poorly correlated among themselves, as well as with general performance measures. Assessments of sportsspecific performance is mostly lacking. Most studies assessing bilateral asymmetries do not actually assess the occurrence of injuries. While injuries tend to accentuate bilateral asymmetries, there is no evidence that pre-existing asymmetries increase injury risk. While training programs reduce certain bilateral asymmetries, there is no evidence that such reductions result in increased sport-specific performance or reduced injury risk.

Conclusions: Bilateral asymmetries are prevalent in sports, do not seem to impair performance, and there is no evidence that suggests that they increase injury risk.